

RatesTools Pipeline Modification Record

Overview

This document summarizes the modifications made to the RatesTools workflow and configuration to improve performance and reproducibility on Cornell BioHPC.

Pipeline launch

Run from the analysis directory using:

```
~/bin/nextflow-25.10.5 run  
/workdir/nina_july2026/silverside_progeny/RatesTools/ratestools.nf -c  
nextflow.config -profile conda -resume
```

Use the local modified `ratestools.nf`, not the GitHub workflow.

Configuration changes

- Converted key paths to absolute paths (reference, libraries, `readDir`, `hc_intervals`).
- Enabled Conda-installed GATK and Picard.
- Added `hc_intervals` parameter (one chromosome per line).
- Added process-specific CPU settings.
- Limited executor CPUs for shared-server use.
- Dam, sire, `chr_file` and RepeatMasker species should be verified before production runs.

Workflow modifications

The original workflow ran one HaplotypeCaller job per sample. It now:

1. Reads chromosome intervals from `hc_intervals.txt`.
2. Runs one HaplotypeCaller per sample × chromosome.
3. Uses a new `gatherSampleGVCFs` process to merge chromosome GVCFs back into one GVCF per sample with GATK `CombineGVCFs`.
4. Performs the original joint genotyping on those reconstructed sample GVCFs.

This reduced variant-calling runtime dramatically by parallelizing across chromosomes.

Caching

Always run with `-resume`. Keep the work directory. Completed alignments, RepeatModeler, GenMap, and chromosome GVCFs can then be reused.

Recommended resources

- `callVariants`: `cpus=1`.
- Limit concurrency with `maxForks`.
- Limit overall executor CPUs to leave resources available for other users.

Lessons learned

`configure.sh` did not correctly populate every parameter. Always inspect the effective configuration with `nextflow config` before a long production run. Keep the modified workflow under version control.